

1. Project Objective

The objective of this project is to enhance a low-contrast image using histogram-based image enhancement techniques. First, the condition of the input image will be analyzed by examining its visual appearance and pixel intensity distribution through its histogram. After that, two enhancement methods, Histogram Equalization and Contrast Limited Adaptive Histogram Equalization (CLAHE), will be implemented to improve the image contrast and make its details more visible. The results produced by both methods will then be compared based on their visual quality and histogram distribution. Finally, the comparison will be used to determine which method provides the better enhancement result for the selected low-contrast image.

2. Problem Identification and Image Analysis

The main problem identified in this project is that the objects in the selected image are difficult to distinguish due to its low contrast. The limited contrast causes the image to appear less clear, making some objects and details difficult to observe. Therefore, before applying any image enhancement technique, the input image needs to be analyzed to understand its initial condition. The analysis includes observing the original image, converting it into a grayscale image, and examining its histogram. The grayscale image allows the intensity information of the image to be analyzed more clearly, while the histogram provides information about the distribution of pixel intensities within the image. By analyzing the histogram, we can identify how the pixel intensities are distributed and determine whether the image has a narrow intensity range, which is an indication of low contrast. This initial analysis will then serve as a basis for applying Histogram Equalization and CLAHE in the next stages.

4. Implementation

The image enhancement process uses two techniques that are based on histograms: Histogram Equalization and Contrast Limited Adaptive Histogram Equalization (CLAHE). First the original image that has contrast is changed into a black and white image so that the numbers that represent the brightness of each pixel can be worked with more easily. Histogram Equalization is then used to make the overall contrast better by spreading out the brightness numbers over a range. Next CLAHE is used to make the contrast better in parts of the image but without making the contrast too strong in any one area. The results from both methods are. Checked against the original image to see how well each one makes the objects and details, in the low-contrast image easier to see.

5. Conclusion

Based on the experiment the low-contrast image was successfully improved with Histogram Equalization and with Contrast Limited Adaptive Histogram Equalization, known as CLAHE. The initial analysis of the image the grayscale image and the histogram revealed that pixel intensities were clustered within a narrow band making some objects and details hard to distinguish. After applying both enhancement techniques the low-contrast image gained contrast and visibility. Histogram Equalization gave a boost to contrast by spreading the pixel intensity values across the whole image. CLAHE on the hand gave a local boost to contrast that adapts to each area and stops any over-enhancement. By comparing the results and the histogram shapes the technique that produced the clearest and most balanced low-contrast image can be chosen as the better method, for enhancing the given low-contrast image.